

- Unlike eukaryotes, prokaryotes can metabolize nitrogen in many different forms. Some can convert atmospheric nitrogen to ammonia, a process called **nitrogen fixation**.
- Prokaryotic cells and even species may cooperate metabolically, including in surface-coating **biofilms**.

? Describe the range of prokaryotic metabolic adaptations.

CONCEPT 27.4

Prokaryotes have radiated into a diverse set of lineages (pp. 583–586)

- Molecular systematics is helping biologists classify prokaryotes and identify new clades.
- Diverse nutritional types are scattered among the major groups of bacteria. The two largest groups are proteobacteria and gram-positive bacteria.
- Some archaea, such as **extreme thermophiles** and **extreme halophiles**, live in extreme environments. Other archaea live in moderate environments such as soils and lakes.

? How have molecular data informed prokaryotic phylogeny?

CONCEPT 27.5

Prokaryotes play crucial roles in the biosphere (pp. 586–587)

- Decomposition by heterotrophic prokaryotes and the synthetic activities of autotrophic and nitrogen-fixing prokaryotes contribute to the recycling of elements in ecosystems.
- Many prokaryotes have a **symbiotic** relationship with a **host**; the relationships between prokaryotes and their hosts range from **mutualism** to **commensalism** to **parasitism**.

? In what ways are prokaryotes key to the survival of many species?

CONCEPT 27.6

Prokaryotes have both beneficial and harmful impacts on humans (pp. 587–591)

- People depend on mutualistic prokaryotes, including hundreds of species that live in our intestines and help digest food.
- Pathogenic bacteria typically cause disease by releasing **exotoxins** or **endotoxins**. Horizontal gene transfer can spread genes associated with virulence to harmless species or strains.
- Resistance to multiple antibiotics has evolved in many pathogenic bacteria, a problem of great concern for public health.
- Prokaryotes can be used in **bioremediation** and production of plastics, vitamins, antibiotics, and other products.

? Describe beneficial and harmful impacts of prokaryotes on humans.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. A process that cannot produce genetic variation in bacterial populations is
(A) transduction. (C) mutation.
(B) conjugation. (D) meiosis.
2. Photoautotrophs use
(A) light as an energy source and CO₂ as a carbon source.
(B) light as an energy source and methane as a carbon source.
(C) N₂ as an energy source and CO₂ as a carbon source.
(D) CO₂ as both an energy source and a carbon source.

3. Which of the following statements is true?
(A) Archaea and bacteria have identical membrane lipids.
(B) The cell walls of archaea lack peptidoglycan.
(C) Prokaryotes have low levels of genetic diversity.
(D) No archaea are capable of using CO₂ to oxidize H₂, releasing methane.
4. Which of the following involves metabolic cooperation among prokaryotic cells?
(A) binary fission (C) biofilms
(B) endospore formation (D) photoautotrophy
5. Which of the following describes a bacterium that lives in the human intestine and causes disease?
(A) commensalist (C) gut mutualist
(B) decomposer (D) symbiotic pathogen
6. Photosynthesis that releases O₂ occurs in
(A) cyanobacteria. (C) gram-positive bacteria.
(B) archaea. (D) chemoautotrophic bacteria.

Levels 3-4: Applying/Analyzing

7. **EVOLUTION CONNECTION** In patients with nonresistant strains of the tuberculosis bacterium, antibiotics can relieve symptoms in a few weeks. However, it takes much longer to halt the infection, and patients may discontinue treatment while bacteria are still present. Explain how this could result in the evolution of drug-resistant pathogens.

Levels 5-6: Evaluating/Creating

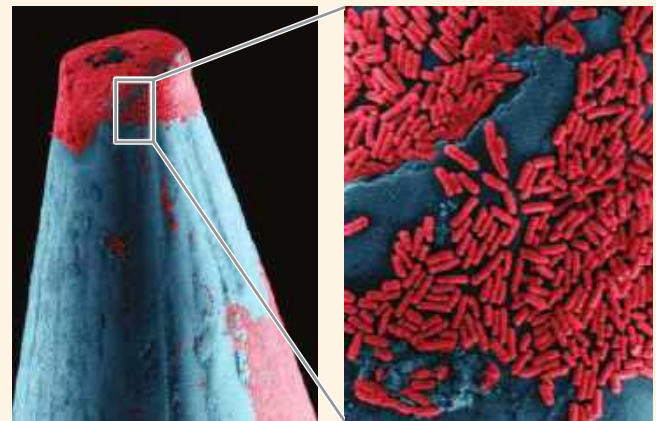
8. **SCIENTIFIC INQUIRY • INTERPRET THE DATA** The nitrogen-fixing bacterium *Rhizobium* infects the roots of some plant species, forming a mutualism in which the bacterium provides nitrogen, and the plant provides carbohydrates. Scientists measured the 12-week growth of one such plant species (*Acacia irrorata*) when infected by six different *Rhizobium* strains. (a) Graph the data. (b) Interpret the graph.

<i>Rhizobium</i> strain	1	2	3	4	5	6
Plant mass (g)	0.91	0.06	1.56	1.72	0.14	1.03

Data from J. J. Burdon et al., Variation in the effectiveness of symbiotic associations between native rhizobia and temperate Australian *Acacia*: within species interactions, *Journal of Applied Ecology* 36:398–408 (1999).

Note: Without *Rhizobium*, after 12 weeks, *Acacia* plants have a mass of about 0.1 g.

9. **WRITE ABOUT A THEME: ENERGY** In a short essay (about 100–150 words), discuss how prokaryotes and other members of hydrothermal vent communities transfer and transform energy.
10. **SYNTHESIZE YOUR KNOWLEDGE**



Explain how the small size and rapid reproduction rate of bacteria (such as the population shown here on the tip of a pin) contribute to their large population sizes and high genetic variation.

For selected answers, see Appendix A.